CHAPTER 30

# Marine degradable plastic and their applications

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## Abstract

Plastic in general is posing great hazard to marine life, more particularly the disposable type of plastic used for various purposes. This has prompted the authorities globally to restrict/ban the use of disposable plastics to protect the nature. Under the circumstances, biodegradable plastics are getting attention as a best replacement for conventional plastics. Biodegradable plastics are getting both importance and controversies due to their definition and their effect on environment as they are expected to gain wider usage. This calls for a deeper understanding about biodegradable plastics and finding scopes for more research where required. This review focuses on information about commonly used biodegradable plastics, the process of their degradation, indicators involved in the process, factors affecting biodegradability and encapsulates all the studies on biodegradation in marine environment. The longevity of any biodegradable plastic depends on its constituents and also on the marine environment. The review also looks into the probable risks such as pollutants, for example, micro/nanoplastics being released, adsorption – desorption of pollutants such as pesticides, the effect they have on biomes and biogeochemical cycles and aims at revealing the possible impact of pollutants on marine environment. Undisciplined and extensive use of biodegradable plastics results in complex effects on marine environment. The discussion finally concludes on the ways of using biodegradable plastics in a way that minimizes negative effect on marine environment and also suggests judicious ways to use it.

## 30.1 Introduction

An estimation says that plastic of about 20 million tons gets deposited in water bodies such as oceans. If this continues, by the year 2050, the weight of plastic in oceans will be more than that of the fish. A majority of floating plastics are conventional high-durable polymers, which get converted into small pieces of 1 cm, or so with time. Therefore, in order to encounter this issue, people have started various measures such as bans,

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fines, and stringent norms over the use of single-use plastics. Carrefour, a multinational retailer from France, levied strict measures for their suppliers, which aimed at ensuring optimal recyclability and packaging post customer use. Though plastic has many negative impacts on environment, removing them completely from circulation also has some socioeconomic impacts (Figs. 30.1–30.3) (Abreo et al., 2018; Abreo, 2018; Abrha et al., 2022; Acock, 2005; Agenda, 2016). To be more specific, sachets are smaller versions of consumer goods, which are convenient for household use. Many daily needs such as cleaners, sanitizers, oils, shampoos, and pickles are marketed in sachets for single use. These sachets are cost-effective, and consumers find it convenient to use them. Purchasing them in big bottles

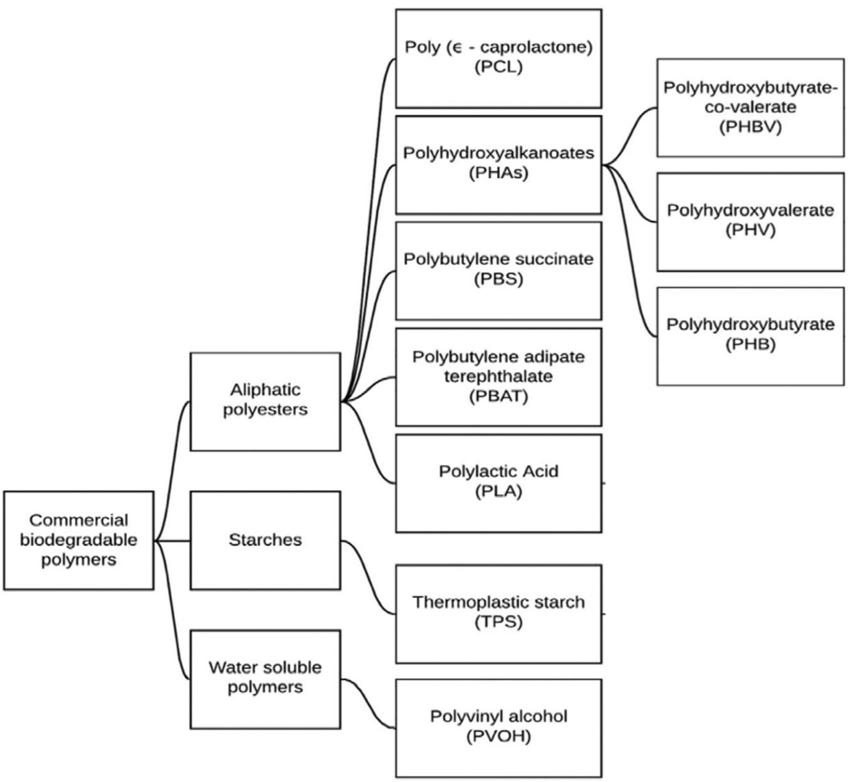


**Figure 30.1** An example of sachets structure (used to supply single dosages of products in multilayered flexible pouches by consumer packaged goods companies) is composed of three layers (inner to outer) PE (polyethylene) heat-sealable layer, vacuum-metalized PET (polyethylene terephthalate) for improved barrier properties, and PET for printing.

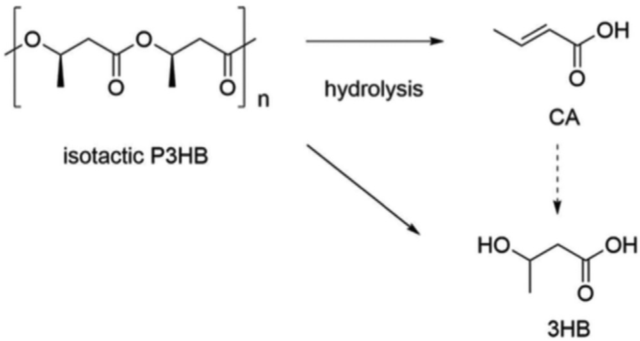


**Figure 30.2** Schematic process of biodegradation of biodegradable plastic.

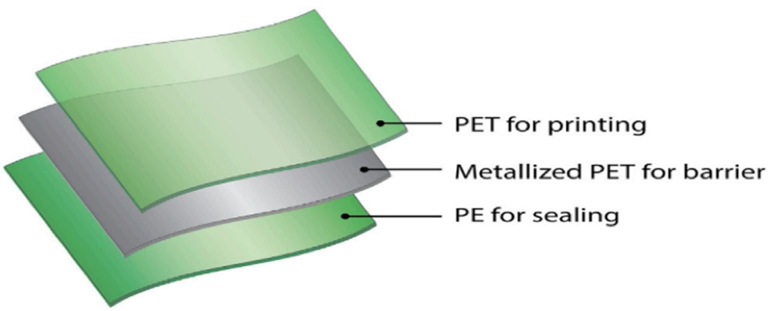


**Figure 30.3** Aliphatic polyesters, polyvinyl alcohol (PVOH), and thermoplastic starch (TPS), biodegradable thermoplastics are in advanced stages of development or commercial usage.

or metal cans means consumer has to spend huge money for storage (Aguirre & D'Esposito, 1999; Akhbarizadeh et al., 2021; Albert & Golledge, 1999).

Sachets can be of single-layer films such as LDPE or in multilayer films such as LDPE, polyethylene terephthalate, and/or polypropylene, Fig. 30.1. The advantage of these materials is that they are economical, lightweight, easy to use, and get adjusted to present packaging converters, but they are not biodegradable and are good for single use and cannot be recycled. Therefore, it is a prime necessary concerned that these single-use plastics have to be removed from the cycle, and they need to be substituted with suitable alternative (Albert & Golledge, 1999; Ang, 2007; Bagheri et al., 2017; Baker & Bednarz, 2003; Baker et al., 2016; Barron & Sparks, 2020; Bednarz & Ludwig, 1997; Avio et al., 2015).

### 30.1.1 Plastic bags and marine pollution

Plastic products are the most commonly found pollutant in oceans of which plastic bags are also found. This information got wide publicity during the year 2014 during which a report about the release of plastic into ocean was released by the International Union for Conservation of Nature and Natural Resources. As per this report, polypropylene, which is used in ropes and nets used for fishing and packaging, contributed to 24% of plastic present in ocean. Polyethylene from straws and plastic covers contributed to 21%. A study was conducted by Ellen MacArthur Foundation about global packaging waste during the year 2016. The results showed that out of the 78 million tons of packaging plastic waste, 32% got released into environments such as forest and ocean (Baker & Bednarz, 2003; Baker et al., 2016; Barron & Sparks, 2020; Bednarz & Ludwig, 1997; Beltrán-Sanahuja et al., 2020; Blacker et al., 2017; Britz & Webb, 2016).

The statistics from World Environment Day Statement, 2018, states that about 500 billion bags made of plastic are used globally per year. With the growth in delivery market, the necessity to use these plastic bags is also increasing. The global pandemic Coronavirus has also increased home delivery, which necessitates plastic use. The common people also feel that single-use containers are more hygienic and safer and protect from infections. Hence, it is becoming difficult to eliminate plastic bags from usage. Though paper bags can substitute plastic bags to a certain extent, they cannot be used to eliminate plastic. That apart, paper bags produce carbon emissions in considerably large quantity, and also, they are not reusable (Britz & Webb, 2016; Burkersroda et al., 2002; Catarci Carteny & Blust, 2021; Chen, 2022; Choe et al., 2021; Cocca et al., 2018; Dauvergne, 2018; Delacuvellerie et al., 2021).

## 30.2 Effects of biodegradable plastics in marine environment

All authorities are trying to restrict use of conventional plastic for which they are updating the policies. Biodegradable plastics (BP) have come up as a fairly good alternative, but there are pros and cons again. There are different opinions about their biodegradability, impact on environment, and environmental significance. Controversies are there about the basic definition about biodegradable plastics. The composition and properties of bioplastics impact its lifespan. The environment inside the ocean also impacts the lifespan. The risks of BP such as emission of pollutants, adsorption, desorption, effect on biomes and biogeochemical cycles, reveal the impacts on the marine environment that may possibly happen. Apart from adding to waste of resources, abundance of microplastic, harmful leachates, and complex effects on natural habitats and environment will negatively impact marine environment, which is the impact of virulent use of BP coupled with inappropriate disposal methods (Dobrucka, 2019; Elagami et al., 2022; Eriksen et al., 2014; Farrell & Nelson, 2013; Fernando, 2015; Filho et al., 2021; Gapuz, 2011).

BP are the polymers that are chemically changed and artificially created, and it can be completely converted into carbon dioxide and water due to biological activity. Microorganisms can convert conventional plastic as BP. Bioplastics and bio-based plastics, though sound similar, are not completely biodegradable. These are plastics made using 100% biological materials or in other compositions, but they are not completely biodegradable. BPs can be created from fossil fuel – based materials. The term BP in general includes bio-based plastics and BPs, but specifically the term represents bio-based plastics (Govil et al., 2021; Haider et al., 2019; Hidalgo-Ruz et al., 2012; KAKEN, 2020; Isensee & Valdes, 2015; ISO 22403, 2020; Jang et al., 2016; Karpušenkaitė & Varžinskas, 2014).

BPs are the main ingredient of plastic products such as nondurable plastics and disposable products such as packing materials, mulching film, and tableware (Abreo, 2018; Blacker et al., 2017). In 2019 these products constituted 49% of plastic production worldwide (Kasuya et al., 1998; Kolybaba et al., 2006; Konti et al., 2022). Before COVID attack, i.e., during 2018, 5% of annual municipal solid waste and 40% of plastic waste (by weight) generated in the United States came from containers and packaging plastics, which is an estimate provided by USEPA (United States Environmental Protection Agency). Only 14.5% of this quantity was recycled. The short service time increased the frequency of their presence in the environment. The presence of disposable or nondurable plastic products in marine litter all around the world is a good example of it.

With the COVID attacking the world, the demand for single-use plastics increased, and plastic bags, food delivery, and electronic packaging were required to use disposable plastic containers to reduce infection and improve hygiene. The pandemic contributed very highly for the increased use of single-use plastic, and the surge was between 250% and 300%. Secondary pollutants such as microplastics increased in the marine environment due to discarded personal protective kits. People are so concerned and used to single-use plastic that even when the pandemic vanishes, this habit will continue for a long time. This also shows that the disposable plastic cannot be taken out from the usage. During the recent times, the ban on usage of single-use plastics imposed by many countries has resulted in increased demand for BPs (Abreo et al., 2018; Abreo, 2018). It is to be noted that BPs constituted just 0.3% in 2019 and 0.6% in 2020; it is expected that the usage of BPs will increase largely soon, the consumption of biodegradable and compostable plastic products was estimated to go beyond 3 lakh tons during 2020, which is almost thrice of what was consumed in 2015 (Kolybaba et al., 2006; Konti et al., 2022; Lott et al., 2020, 2021; Lucas et al., 2008; Meereboer et al., 2020; Mozaffari & Kholdebarin, 2019; Muhamad et al., 2006).

The most frequently asked question about BPs is whether it gets degraded complete in real and complex natural circumstances and marine environment. The BPs are conditionally biodegradable due to their different rate of biodegradability under aerobic or industrial anaerobic conditions. BPs can be divided into different categories such as industrial compostable, soil biodegradable, and marine biodegradable as per the retrieval of American Standard for Testing and Materials (ASTM) standards of test methods. These are categories certified by the European certificate scheme. There is one more standard system for BPs, the International Organization for Standardization (ISO) system. Most of the certified BP can be biodegraded in aerobic or industrial anaerobic conditions.

High temperature and moisture in right proportion act as catalyst in biodegradation for industrial composting. It should be noted that microbes present in aquatic environment are lesser compared with compost and soil. This results in higher composting of BPs in industrial composting, which is estimated to be 72.3% over 75 days. But composting is much slower in aquatic environment, which is estimated to be 47.1% in over 155 days. It should also be noted that the laboratory conditions and natural environmental conditions vary to a considerable extent. The material density of majority of BPs is more than that of surface water, which includes seawater (SW). Upon entering the marine environment, they will get down to the bottom as the bottom of the sea is dark due to less penetration of light and also has anaerobic, anoxic state combined with low temperature. This causes deviations and fluctuations between supposed biodegradation and real biodegradation rate in the environment.

## 30.3 Biodegradation ability in the marine environment and related factors

### 30.3.1 Biodegradation process and description indicators

- Biodegradation can be split into three stages such as fragmentation, hydrolysis, and finally, biodegradation of oligomers and monomers. Biodeterioration is the first step wherein processes dominated by biotic and abiotic factors take place. This will initiate fragmentation along with other abiotic factors such as waves scouring, which causes mechanical stress, sand abrasion, chemical factors, animals all contribute toward fragmentation.
- Hydrolysis, as a depolymerization reaction, is the rate-limiting step in the biodegradation of BPs. The rate of hydrolysis of entire BP can be almost uniform (bulk erosion mechanism) or may be determined by the process on the surface (surface erosion mechanism). BPs generally are aliphatic polyesters or contain glycosidic bonds. The most common BPs are PLA and starch-based materials. They get hydrolyzed by extracellular enzymes. Some of them will get hydrolyzed without enzymes in alkaline environment. But it is not easy for most of the hydrolases to permeate the structure of BPs, and hence, bulk hydrolysis rates of BPs are determined by bulk diffusion rate and migration distance of water molecules.
- The final steps are the degradation of hydrolysis products. The products can be taken in after which it can be mineralized by microbial cells. Hence, it can be bifurcated as bioassimilation and mineralization (Delacuvellerie et al., 2021; Desai, 2020; Dobrucka, 2019; Elagami et al., 2022; Eriksen et al., 2014). This entire process need not take place inside the microorganism. Anaerobic microorganisms can degrade BPs to methane and then finally mineralize them in vitro. Some people believe that the first step of biodegradation is the deterioration process when they start to lose their physical and structural chemicals of simple compounds. The process of deterioration will not change the environmental behavior of BPs significantly (Farrell & Nelson, 2013).

### 30.3.2 Factors affecting biodegradability

There is no environmental significance regarding considering this process as part of biodegradation. The three steps involved in biodegradation may not necessarily happen in the same sequence. Like fragmentation, which is the first step, can be the outcome of hydrolysis, which is the second step. Keeping this in mind, some consider only two steps to be involved in biodegradation, which are depolymerization and utilization of by-products. Fragmentation is considered as simultaneous biotic or abiotic process. Therefore, the result of the previous one can speed up the further step to a great extent. Fragmentation of BPs can speed up the hydrolysis reaction, more so when the surface erosion plays an important role. Many indicators are used to describe the biodegradation of BPs such as weight loss, molecular weight loss, lowering in biochemical oxygen demand and carbon dioxide produced. The entire biodegradation process cannot be described with one indicator. However, weight loss is the main and largely used indicator despite the fact that it cannot track the biodegradation of secondary microplastics evolved during the BP fragmentation as they disappear fast in the environment.

Photogrammetric method is often used to get an approximate weight loss. Weight loss is a continuous process during biodegradation, which gets influenced by many factors such as release of soluble additives and loss of fragments. Chain shortening induces molecular weight reduction, which is an indicator for the hydrolysis of BPs as considerable molecular weight loss happens only during hydrolysis (Nadkarni & Nadkarni, 2007; Nanda et al., 2022; Narancic et al., 2018).

The hydrophobicity of a plastic surface may provide them with varying microbial compositions from those on glass, ceramics, or any other natural environment. In general, it is seen that the adhesion between bacteria and hydrophobic surface is greater than that of hydrophilic one. The size, roughness, and shape affect the specific surface area of the plastics. Microorganisms take more time to get in contact with the plastics with small areas. The specific surface area of BPs increases in the biodegradation process, with change in roughness and decrease in size (Norén, 2007; Nurhati & Cordova, 2020).

#### *30.3.2.1 Hydrolysis*

The hydrolysis rate of the BPs majorly impacts the biodegradability. A few important properties are hydrolyzable bonds and neighboring groups. Typically, hydrolyzable bonds and neighboring groups determine the kinetics of bond cleavage. Polymers based on cellulose, starch, and proteins exhibit better marine degradability rates compared with polyesters, which has ester bonds hydrolyzable and could reinforce the biodegradability with joined with polyesters.

The biodegradability of cellulose acetate formed due to cellulose acetylation reduces with the increased degree of substitution. The cause for this might be hindrance of carboxyl bonds during the hydrolysis. Yet another example of neighboring group impact is polyethylene terephthalate or polyester (PET) as these materials are persistent in oceanic atmosphere due to aromatic rings adjacent to ester bonds (Desai, 2020; Meereboer et al., 2020; Rai et al., 2021).

#### 30.3.2.2 Additives

These are the chemical substances used to provide specific quality to the plastic. Additives are complex and diverse and are not disclosed and guarded as trade secret. The most important additives of BPs are crystalline clearing agents, antihydrolysis stabilizers, melt enhancers, chain extenders, plasticizers, and so on. Inorganic additives may cause degradation of plastic products as they are not compatible (Albert & Golledge, 1999).

#### 30.3.2.3 Chain extenders

Chain extenders play the role of increasing the molecular weight and enhance cross-linking during production, which in turn prolongs the biodegradation time (Britz & Webb, 2016). Chain extenders and antifouling agents can inhibit extracellular enzyme activity or prevent the microorganism from producing enzymes. Antihydrolysis stabilizers for condensed BPs can improve their aging stability considerably. Hence, the leaching of plasticizers makes the plastic brittle and stimulates fragmentation. Few other additives such as photostabilizers and colorants can cast effect on the thermal and UV stability of plastics, which may have effect on fragmentation and hydrolysis of BPs.

#### 30.3.2.4 Natural environmental condition

This includes habitats and climatic zones, which can lead to differences in the degradation of BPs. In aquatic environment, environmental factors are light, salinity, heat, pH, electron acceptors, nutrients, grain size of sediments, and population of microbes.

All these factors affect microbial composition, abundance, and activity along with abiotic effect on degradation.

#### 30.3.2.5 Fragmentation

Fragmentation of BPs gets accelerated by UV light and slowly produces low-molecule-weight oxidation products for bioassimilation, but the UV light effect gets attenuated rapidly with depth of water. Light may also decelerate biodegradation, e.g., cross-links that are formed during photoirradiation-induced reactions can slow down biodegradation. Light has the capacity to moderation the proportion of photosynthetic autotrophs (Desai, 2020).

Specific microorganisms are required for the enzymatic hydrolysis of BPs, and these microorganisms should secret extracellular enzymes (hydrolases) as well as oxidation reactions. On the contrary, the microorganisms that could be involved in bioassimilation are extensive, so the demand for microorganisms in this step is more likely to be satisfied in the natural environment. Furthermore, inhibiting other microorganisms, which are not a part of biodegradation, may accelerate the biodegradation of BPs, since a broad-spectrum fungicide was found to accelerate the biodegraded rate. Classically, hydrolases effective in the hydrolysis of biodegradable polyesters (PLA, PCL, and PBS) belong to cutinases (can hydrolyze ester bonds in cutin), lipases (can hydrolyze ester bonds in lipids), and polyhydroxyalkanoates (can hydrolyze ester bonds in carboxyloxoesters of bacterial polyesters); they have common features, such as extensive substrate-binding grooves for macromolecular polyesters, catalytic amino acid triad in active sites for binding subtract, and not substrate-specific. Some other hydrolases that can be used for biodegradation include amidases and acetyl esterases. In addition, physical shearing caused by currents/waves, wind, fauna, or sediment (especially sands) improves the fragmentation of BPs (Abrha et al., 2022; Dobrucka, 2019).

## 30.4 History of standardization of marine degradability evaluation methods

With growing awareness about the problem of plastic waste in aquatic environment, the G7 (Group of seven) adopted the Ocean Plastics Charter in June 2018 in the summit that took place in Canada. The document specified how this problem countered and what measure needs to be taken country-wise. Countries such as Canada, Germany, France, United Kingdom, and Italy signed the charter while the United States and Japan remained out. Validation tests were conducted for technology to evaluate the marine degradability of plastics under OpenBIO during the period 2013–16. The results arrived at were made use to develop a manual for establishing fresh standards.

OpenBIO, as the name suggests, aims to develop a differentiation strategy to expand the bioplastics market. HYDRA, the German marine research institute, took the lead of the project along with Novamont of Italy, which is a bioplastics manufacturing unit, and BASA of Germany. A similar international standard is "OK biodegradation MARINE label" based on ASTM D7081 and D6691, which is certification of marine BP. But both the underlying ASTM standards were removed in 2015 as being inadequate as a method for evaluating marine degradable plastics based on OpenBIO validation results. In view of the aforementioned, international standardization of evaluation methods related to aquatic degradable plastics is being led by the EU from data acquisition to establishment of standards. Japan took the lead in standardizing the compost and soil degradability of bioplastics and biomass plastics in the past, which led to increase of Japanese products in the market. However, Japan was unable to find the significance of marine degradability because domestic recycling systems were established in advance. This led Japan to a wait-and-see stand to the EU's move to establish standards (Akhbarizadeh et al., 2021; Bagheri et al., 2017; Baker et al., 2016; Bednarz & Ludwig, 1997; Chen, 2022).

## 30.5 Details of the new ISO 22403 and 22766 standards for marine degradable plastics

Marine Degradability of Plastics and Positioning of the New Standards: The breakdown of plastics in aquatic environment arises through the complex effects of many combined factors. Thus, it is held that the environmental impact of plastics exposed to the marine environment should be evaluated comprehensively based on following three main aspects:

- Field evaluation to be done according to the environment in which waste plastic is actually exposed such as composition of SW, water depth, concentration of microorganisms, and changes in water temperature and sunlight due to weather factors.
- Biodegradability evaluation in a laboratory environment managed artificially to have the same conditions.
- Safety evaluation as a chemical substance ISO 22766 specifies field testing methods.

ISO 22403 specifies the testing method in the laboratory environment in and the criteria for determining biodegradability in the ocean (determination based on data obtained in a laboratory environment). For the safety evaluation in application of the ecotoxicity evaluation method OECD

(Organisation for Economic Co-operation and Development) is recommended (Mozaffari & Kholdebarin, 2019; Nadkarni & Nadkarni, 2007; Nanda et al., 2022; Narancic et al., 2018; Norén, 2007; Nurhati & Cordova, 2020).

### 30.5.1 Overview of ISO 22766: 2020 for marine degradable plastics

The method of evaluation mentioned in ISO 22766 is known as a disintegration test, and it varies from biodegradation evaluation. Disintegration test is a field evaluation, which is carried out in coastal area, i.e., littoral and from coastline to a depth of 200 m, i.e., sublittoral zone, where maximum waste plastic gets collected. Three different types of plastic are placed in the test area. A good reference for testing the degree of disintegration (for example, Kaneka 's PHBH), a reference with a poor degree of disintegration (low-density PE), and the plastic film to be evaluated. The degree of disintegration is measured by the remaining area of the film. The period will be up to a maximum of 3 years, but the test is closed if the degree of disintegration reaches 90% or more; in other words, the remaining area is less than 10% of the original film.

### 30.5.2 Overview of ISO 22403: 2020 for marine degradable plastics

ISO 22766 describes the method for evaluating the degree of disintegration in sea, ISO 22403 describes the method of evaluation for biodegradability in the seas. Biodegradability is considered as the plastic getting diffused into carbon dioxide and water by microorganisms present in the ocean. Hence, biodegradability is evaluated by measuring the carbon dioxide released by biodegradation when a sample in the form of powder is sealed in with marine water and sediment. The performance requirement of ocean biodegradability was 90% or more of the carbon contained to be broken down into carbon dioxide within a period of 24 months or having better biodegradability than cellulose, which is the main ingredient of plant fiber.

### 30.5.3 Impact of both standards on the development of marine degradable plastics

It is seen that determination of evaluations on required standards takes a testing period of about 2–3 years. There are no shorter period tests available as on date. Due to these criteria, plastics that are already in use as reference samples when the marine degradability standard was reviewed only are eligible to enter the market (Fig. 30.1). On the contrary, it can be an obstruction for the plastics developed later to enter the market if the same verification is required (Fig. 30.3) (Barron & Sparks, 2020; Peake, 2020; Peng et al., 2022; National Research Council (U.S.) & Committee on Polymer Science and Engineering, 1994; Posadas, 2014).

### 30.5.4 Polylactide

Polylactide or PLA is the most widely used biodegradable plastic. This degrades in the compost with time proved a dramatically reduced degradability in SW, similar to it in pure water. A study was conducted by Tsuji and Suzuyoshi about the degradation properties of PLA films of 0.05 mm thickness in aquatic water in natural conditions and in the laboratories in SW, i.e., under static conditions. The results showed that the overall properties of PLA films remained unchanged even after 10 weeks under lab conditions.

Due to plasticization process, tensile strength and Young's modulus increased marginally during the initial stages of experiment under natural conditions in the ocean, fracture of the films was the result of mechanical forces, which took place after 5 weeks. This resulted in more weight loss and mechanical reduction, but GPC did not show any significant changes in molar mass. Degradation of PLA splines of 4 mm thickness was studied by researchers in Lorient harbor of France during 6 months. It was observed that there was no significant change in molar mass and mechanical properties excluding some tensile strength loss (Barron & Sparks, 2020). The enhanced salinity in SW as against the distilled water at room temperature affects the diffusion of water into polyester, impacting the degradation rate in SW; it becomes slower than in pure water. To further predict the lifetime of PLA splines of 2 mm thickness in marine water, scientists have extended the degradation time to 1 year in natural SW. The experiment affirmed the earlier data, which showed that PLA was not easily degraded in marine water.

### 30.5.5 Polyhydroxyalkanoates

PHAs unlike PLA undergo hydrolysis faster in sea water also. Already in 1992, seminal studies on the degradation of PHAs in SW have been reported. It was found by the authors that the PHA degradation mechanism in SW followed surface erosion, as was reported in soil and compost.

But the degradation speed of PHAs in marine water was slow significantly. Researchers have reported that PHBV films of 0.115 mm thickness, which degraded completely in compost within 6 weeks, only proved 60% weight loss when immersed in marine water after the same time. The blend of PHBV with 60 wt.% PHB resulted in a weight loss of 100% and 38% as degraded in compost and SW, respectively. Degradation of PHB and P (3HB-3HV) films was studied by researchers in South China ocean for a period of 160 days. Several strains of PHA-degrading *Enterobacter* sp. were identified along with *Bacillus* sp. and *Gracilibacillus* species. They also proved that the shape and size of the material played a significant role in degradation process. The study of PHB and P(3HB-*co*-3HV) films in SW after 160 days showed that the weight loss of film with 0.1 mm thickness was 38% and 13%, respectively, implying that degradation of PHB was faster than that of P(3HB-*co*-3HV) though the conditions were same. Furthermore, the weight loss increased to 58% and 54%, respectively, when the thickness of the film was reduced to 0.005 mm. In surface corrosion process, the size of the sample, particularly the thickness and surface area determined the rate of degradation. Recent works from scientists have investigated about the rate of biodegradation of PHA in the marine environment and applied this to the lifetime estimation of PHA products (Quoquab & Mohammad, 2020; Rai et al., 2021; "Research and Library Service Briefing Note: Comparison of Environmental Impact of Plastic, Paper and Cloth Bags," 2018; "Seibunkaisei kino wo ON OFF," 2020; Tachibana et al., 2013; Taniguchi et al., 2019).

### 30.5.6 Polycaprolactone

Many papers have reported the degradation of PCL in ocean water. Researchers have studied the degradation properties of PCL in lake, river, and sea waters and proved that PCL can degrade in most of the natural waters due to the presence of microorganisms, which can degrade in different waters. They have also done the evaluation of degradation of PCL fibers after 12 months soaking in deep SWs at Rausu, Toyama, and Kume, respectively, in Japan. It's a noticeable fact that with noticeable mechanical decline, the surface of PCL fibers appeared with heterogeneous pinholes and cracks under the action of microorganisms, which suggests that significant biodegradation of the fibers occurred in the waters. Five PCL-degrading bacteria were screened and separated from deep SWs pumped up at those three locations by using a PCL granule-containing agar medium.

The isolates were found to belong to the genus *Pseudomonas*, which is very widely distributed in natural environments and includes several aliphatic polyester-degrading bacterias, as well as two others: *Alcanivorax* and *Tenacibaculum*, which had not been reported as aliphatic polyester-degrading bacteria. A laboratory test reported fast degradation of PCL films with 0.1 mm thickness when immersed in SW from the bay. The degradation properties were found to be similar to that of PHBV in the same SW from the bay but reduced in the water from the ocean. The rate of degradation reduced as the thickness of the material increased, the 0.05 mm thickness PCL films reduced the mechanical strength by 100% and lost 30% of its original weight in natural SW after 5 weeks. In our recent work, 2 mm thickness PCL splines were dipped in natural SW located in BoHai China, and a 30 wt.% loss compared with its original weight after 52 weeks was measured. Analysis of the remainders with respect to size, its molecular weight, and mechanical strength indicated that a surface erosion mechanism occurred, similar to other reports from the literature (Agenda, 2016; Govil et al., 2021; Lott et al., 2020; "Seibunkaisei kino wo ON OFF," 2020; Thellen et al., 2008; Thevenon et al., 2014; Thompson et al., 2004).

## 30.6 Promising seawater-degradable polymers without the need for blending or modification

Some polymers are SW degradable for which microbial degraders are widely distributed in marine ecosystem. Starch or cellulose-based polymeric materials get degraded readily in SW because microbial degraders are present in different natural environments. But in majority of cases, polymers are used as main content of the material, instead they are used only as fillers. Also, some polyesters such as PCL and PHA are said to degrade in water at a very slow pace compared with degradation in soil. It is required to mention new aspects of chemical routes to PHB here: besides bacterial poly[(R)-3-hydroxybutyrate], P[(R)-3HB], a perfectly stereoregular, pure isotactic crystalline thermoplastic material, also cyclic lactones (4-membered β-butyrolactone or more recently eight-membered cyclic diolide) can be conducted to obtain P3HBs with adjustable tacticity and properties, which also control the degradation rates. More recently, in the context of chemical recycling, for the abiotic hydrolysis of poly[(R)-3HB] under acidic or basic conditions, Yu and coworkers reported the formation of two monomeric hydrolytic products, 3HB and CA (Fig. 30.4). 3HB and CA were detected as major hydrolytic products

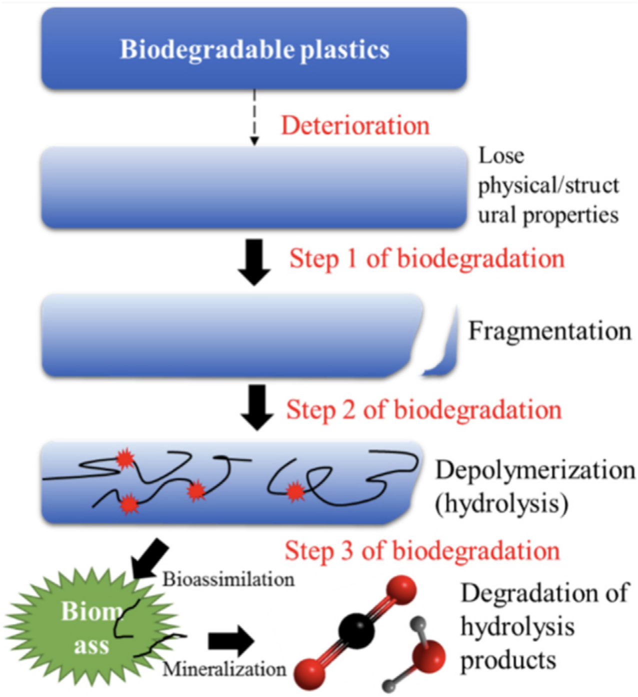


**Figure 30.4** Hydrolysis of P3HB to 3HB and crotonic acide (CA).

from alkaline hydrolysis, while poly[(R)-3HB] was tolerant to hydrolysis under moderate acidic conditions. On the contrary, poly[(R)-3HB] was completely degraded into 2% of 3HB and 90% of CA in concentrated acid, during which 3HB was dehydrated to CA. Accumulation of two monomeric products, 3HB and CA, was in compliance with the mass loss of P3HB films, indicating a sequential degradation from the ends of P3HB polymers and oligomers. This shows the potential of P3HB to be suitable SW-degradable polymer also by synthetic approaches; its degradation product 3HB can also be metabolized (Fig. 30.4) (Volova et al., 2010; Wang et al., 2021; Wang et al., 2022; Wang, Huang, Zhang et al., 2020; Wang, Huang, Zhang, Li, et al., 2020).

Further, polyesters such as PGA and PLGA, which have quick abiotic hydrolysis rates, can be applied in SW-degradable minerals. PGA, which synthesized by the ROP of glycolide (GA), is the simplest of aliphatic polyester, which has poor solubility, high melting temperature ($T_m$) (225°C−230°C) and great mechanical strength (Young's modulus $E = 12.8$ GPa). PGA hydrolyzes in pure water with considerable loss of mass within 5 months, and tensile strength is lost after 1 or 2 months. Random copolymerization of GA units within LA segments disrupts the regularity of the polymer and reduces the crystallinity of copolymers PLGA. Thus, the amorphous copolymer PLGA exhibits improved toughness and solubility along with accelerated and adjustable biodegradability caused by the more available ester bonds. Researchers have studied the degradation of PLGA in both artificial SW and freshwater at 25°C under fluorescence light (16 hours light and 8 hours dark) for 1 year. PLGA got completely degraded in SW after 270 days as determined by time-dependent weight loss and change in molar mass; similar results were obtained for freshwater. Moreover, the initial PLGA evaluated with a unimodal molar mass curve in GPC. PLGA incubated in freshwater or seawater exhibited broadening in GPC for partially degraded samples showing chain scission and hydrolysis. The decrease in molar mass indicated bulk degradation. The amorphous nature of the polymer might be responsible for the faster hydrolysis and complete degradation of PLGA, facilitating the diffusion of water throughout the bulk. Although the specific degradation process and the final product of PLGA in SW are yet to be studied, the quick biotic hydrolysis and the produced small molecular oligomers will increase the chance of the assimilation by microbes. It should be noted that commercial PGA and PLGA are synthesized by ring-opening polymerization from high-cost LA and GA intermediates to ensure the high molecular weight, narrow molecular weight distribution, high purity, and low by-product for medical requirements. The high costs limit their use in other than biomedical applications. PET, one of the most important synthetic polymers used today, is also a good SW-degradable polymer in SW. While efficient PET-ase was artificially produced, a few species of bacteria and fungi in marine water have been described as capable of partially degrading PET to oligomers or even monomers. However, it is to be noted that all known PET hydrolases have relatively low turnover rates. Also new polyesters, which might undergo SW degradation, can be potential replacement for the slowly degrading PET. In a recent study, copolymers based on γ-butyrolactone and *trans*-hexahydrophthalide were prepared, which exhibited comparable barrier and mechanical properties to petroleum-based PET and superior to bio-based PLLA. Such copolymers can be used in packaging applications with a closed loop lifecycle through either their degradability or chemical recyclability; however, no SW degradation has been reported till date (Thevenon et al., 2014; Thompson et al., 2004; Tsuji & Suzuyoshi, 2002; Van Rossum, 2021; Viera et al., 2021; Volova et al., 2010).

Apart from hydrophobic polymers, water-soluble polymers are a good option as SW-degradable polymers. Polyvinyl alcohol or PVA or polyethylene glycol or PEG are good examples for this. PVA is manufactured in large scale by hydrolysis of polyvinyl acetate. The water solubility of PVA is highly dependent on its alcoholysis degree and the molecular weight. Different grades of PVA with varied mechanical properties and water solubility have found their applications in industries such as food packaging, cosmetics, coating, textile, and paper. Though PVA can be biodegraded under both aerobic and anaerobic conditions and a variety of microorganisms (e.g., members of the genera *Pseudomonas, Sphingomonas* and *Sphingopyxis*), the distribution and number of microorganisms with the ability to degrade PVA in SW are lesser than those of the ones that can degrade aliphatic polyesters. Microorganisms capable of degrading PVA are not ubiquitous but are distributed in specific environments such as wastewater, particularly the water discharged from textile and paper mills containing PVA.

Researchers have searched for PVA-degrading microorganisms in SW and isolated new strains of the genus *Thalassospira*. The genus *Thalassospira* comprises Gram-negative, halophilic bacteria belonging to the family *Rhodospirillaceae* of the class *Alphaproteobacteria* and includes polycyclic aromatic hydrocarbon-degrading species that are found in oceans, waste-oil pools, and petroleum-contaminated SW. Although microorganisms being able to assimilate PVA are rarely found in marine environments, there is still a chance of PVA degradation in these specific environments. On the contrary, for ethylene vinyl alcohol copolymers, weight loss and $CO_2$ evolution measurements were conducted in respirometric tests carried out in the presence of marine sediments and selected marine microorganisms. This can noted that no significant degradation was reported; however, a marine actinomycete was found, which was able to grow in the presence of EVOH as single carbon source (at a rate too slow to be detected by the respirometric measurement) (Akhbarizadeh et al., 2021; Beltrán-Sanahuja et al., 2020; Cocca et al., 2018; do Val Siqueira et al., 2021; Mozaffari & Kholdebarin, 2019; Muhamad et al., 2006; Rai et al., 2021; "Research and Library Service Briefing Note: Comparison of Environmental Impact of Plastic, Paper and Cloth Bags," 2018; Tsuji & Suzuyoshi, 2002; Wang et al., 2022; Weber et al., 2018; Wei et al., 2021).

## 30.7 Seawater-degradable blends containing degradation promoters

### 30.7.1 Blends with readily degradable polymers

To improve degradation rates in SW, addition of readily biodegradable fillers can help to accelerate to degradation of the polymer matrix. This strategy was widely applied and investigated when some products are required to give a certain degree of biodegradability in soil or compost (Abreo, 2018). Starch and cellulose (or derivatives) have been used in this approach, as they are economical, easily available, exhibit good compatibility, and can be readily degraded by yeast, fungi, and various bacteria. Researchers have investigated the effect of blending starch (5% and 8%) into polyethylene on the degradation in the Baltic Sea at Nordic Wharf of Gdynia harbor (Cocca et al., 2018). It was expected that the degradation process of PE should be sped up through microbiological consumption of the starch particles, producing a higher surface/volume ratio of the polyethylene matrix. The experiment took place for 20 months, and it was found that in natural SW, the enzymatic hydrolysis of starch occurred and demonstrated by clear erosion of the surface and weight loss, but the remaining PE with the increased surface area did not show increased weight loss compared with bulk PE control (less than 1% weight loss was determined). This result demonstrated that blending polyethylene with starch (5% and 8%) did not increase the microbial degradation of PE bulk in SW under natural conditions.

Testing the degradation of various proportions of PLA starch $CaCO_3$ − glycerol blends in seawater under lab conditions was done by researchers. After incubating for 28 days, samples were fragmented with a maximum weight loss of 73%, similar to the removal of additive amount (72%). The degradation was not complete even after 4 months under these conditions, the degradation of PBAT and its composites containing starch (PBAT-starch) proved similar results (Haider et al., 2019). The test splines were immersed in static river water, static SW, natural SW, static sterilized distilled water, static sterilized SW, and static sterilized lab-prepared SW for 56 weeks. The results underlined that the degradation of pure PBAT was slow in all water samples with a maximum weight loss of only 4.7% after 56 weeks.

Recently, BioLogiQ, a US biofuels manufacturer, announced the successful development of a new thermoplastic polymer, named NuPlastiQ, which is based on starch extracted from potato. A combination of NuPlastiQ and PBAT is called a NuPlastiQ MB biopolymer. The new product so obtained is said to undergo marine biodegradation within 1 year. However, the degradation properties, final products, and degradation properties of different shapes are to be studied further. The degradation rates of blends of similar starch increased dramatically when the polymer matrix itself proved to be degradable in SW. Researchers have investigated the degradation of PHBV blended with corn starch at four different places in coastal water southwest of Puerto Rico for 1 year. Microbial enumeration at the four stations revealed considerable flux in the populations over the year. However, in general, the overall population densities of 870 CFU/mL at the deeper-water station were one or two orders of magnitude less than that at the other stations. Starch degraders were 10 − 50 times more prevalent than PHBV degraders at all of the stations. Biphasic weight loss was observed for the starch − PHBV blends at all investigated places, as both starch and PHBV are known to undergo biodegradation in SW. However, the degradation rates of the polymer blend, as determined by weight loss and deterioration of tensile properties, correlated with the amount of starch present (100% starch > 50% starch > 30% starch > 100% PHBV). The additional incorporation of PEG into the mixture slightly retarded the rate of degradation, probably as PEG enhanced the adherence of the starch granules to PHBV in the blends. The weight loss rate of starch from the 100% starch samples was about 2% day −1, while the loss rate of PHBV from the 100% PHBV samples was about 0.1% per day. A predictive mathematical model for loss of individual polymers from a 30% starch − 70% PHBV formulation was developed and experimentally validated. The model showed that PHBV degradation was delayed 50 days until more than 80% of the starch was consumed and predicted that starch and PHBV in the blend had half-lives of 19 and 158 days, respectively. The degradation of few starch − polyester blends in SW increased when compared with the pure polyesters, the overall degradation rates are much slower in comparison with degradation in soil or compost. As the polyester matrices exhibit generally a low SW degradability, the weight loss of the composite after immersing in SW is mainly due to weight loss of the fillers. Apart from this, the removal of the fillers results in an accelerated loss of the mechanical properties of the material and speed up their fragmentation process. For biodegradable polyester matrix such as PHB, that supposed to be helpful to accelerate their end of life by microbe; for nondegradable polymers such as PE, that may lead to their fragmentation before significant biodegradation can occur and might accelerate the generation of microplastics (Fernando, 2015; Lott et al., 2021; Narancic et al., 2018; Peake, 2020; Rai et al., 2021; "Research and Library Service Briefing Note: Comparison of Environmental Impact of Plastic, Paper and Cloth Bags," 2018).

### 30.7.2 Water-soluble degradation promoters

PVA was blended with readily degradable substances such as biopolymers or biodegradable polyesters to find out the promotion of the degradation process. Starch [or thermoplastic starch (TPS)]-PVA blends are of particular interest as they are highly compatible. TPS and PVA can be blended in different ratios to customize the mechanical properties and degradability of the material for a variety of applications. The rate and extent of biodegradation of the blends were estimated from the biodegradation behavior of the individual components. Deviations have been reported as the degradation mechanisms can interact with each other, for example, the less degradable phase can decrease (or inhibit) the access of water into the other (normally more degradable) component. Therefore, the rate and extent of degradation of the blend may not be directly predictable from the extent and rate of degradation of the components. A systematic study on the anaerobic degradability of a series of TPS-PVA blends was conducted to determine their fate upon disposal in either anaerobic digesters or bioreactor landfills. It was found that the presence of starch in the mixture had a positive influence on the rate of degradation of PVA solely; concurrently, a significant reduction was witnessed in the overall period of degradation. According to these observations, starch can be considered more than just a filler in a polymer mixture, as it even contributes to inducing the biodegradation of PVA (do Val Siqueira et al., 2021; Tsuji & Suzuyoshi, 2002; Wang, Huang, Zhang et al., 2020; Wang, Huang, Zhang, Li, et al., 2020; Wei et al., 2021).

When blended with polyesters such as PHB, which have faster degradation rate, both components degrade independently as water uptake rates are similar and different microorganisms degrade the components of the polymer blend. PVA also increased the mineralization rate of the PHB fraction in the blends. The accelerating effect of the PVA seemed to be related to processing and the lowering of PHB crystallinity in polyester-rich blends (Hidalgo-Ruz et al., 2012). The primary investigation of the influence of PVA on biodegradation of PCL in the PVA/PCL blend films was carried out by BOD measurements. The experiment was carried out in axenic cultures of a specific PCL assimilating actinomycete that was isolated from the compost deriving from the organic fraction of household waste. The biodegradation of pure PCL film was completely suppressed when the cultures were supplemented with PVA, even though PCL mineralization was already started. It looks like PVA tended to strongly adsorb on the PCL surface when it was added to the aqueous culture medium, resulted in a change in the surface properties of PCL films, which substantially depressed the hydrophobicity required for guaranteeing the accessibility of the polymer bulk to esterase enzymes and strongly inhibited their propensity toward biodegradation. The aforementioned studies are based on PVA blends in compost, certain bacterial culture, or other freshwater bodies, the degradation properties of PVA blends in SW may differ from them based on the variations in environmental factors. Unfortunately, there is only very little research done in this area. Researchers have reported the biodegradation of PVA/LLDPE plastic film by a consortium of marine benthic vibrios. Marine bacteria, *Vibrio alginolyticus* and *Vibrio parahaemolyticus* from sediments, were evaluated for their ability as a consortium to degrade blends of poly(vinyl alcohol)/LLDPE films over 15 weeks (shaken at 120 rpm). Well-engineered and manufactured PVA can be used as an effective hydrolysis accelerator for the PCL matrix. The channels left by the quick dissolution of PVA facilitate the entry of water and microorganisms into the materials to contact the PCL, thereby promoting the biodegradation process of the PCL matrix itself (Elagami et al., 2022; Jang et al., 2016; Lott et al., 2021; Nadkarni & Nadkarni, 2007; Peng et al., 2022; National Research Council (U.S.) & Committee on Polymer Science and Engineering, 1994; "Research and Library Service Briefing Note: Comparison of Environmental Impact of Plastic, Paper and Cloth Bags," 2018; Thellen et al., 2008; Thevenon et al., 2014; Thompson et al., 2004). On the one hand, the biodegradable material is given a certain degree of water solubility, which makes the material dissolve or disintegrate quickly, resulting in the loss of structure and performance in a short and controllable time. On the other hand, the water solubility of the material increases the probability of the material contacting with water and microorganisms, thus promoting the hydrolysis process and achieving rapid degradation in seawater (Choe et al., 2021; Isensee & Valdes, 2015; National Research Council (U.S.) & Committee on Polymer Science and Engineering, 1994).

## 30.8 Future scope in marine degradable plastics

Reducing the use of plastic may not be possible soon, and it may take a long time to implement that. With the present methods of plastic disposal, the buoyant plastics in the ocean are going to be four times more by the year 2050. Remarkable progress is achieved in understanding the plastisphere and its effects on the environment, but many more important facts need to be sorted. More studies need to be carried out to evolve eco-friendly plastics, which are biodegradable, oxodegradable, or compostable and should be plant-based so that users and concerned authorities can take the right decision. Along with this, future studies about plastic biodegradation should also include a study on enzymes that are responsible for the breakage of plastics and should implement multiomics method of deciphering biochemical mechanisms of plastic biodegradation to show a perfect proof of biodegradation.

There are many other questions such as the effect of plastisphere on coastal habitats such as seagrass meadows, mangrove forests, or coral reefs. The ecotoxicological effect and pathogenicity of plastisphere on wildlife need to be assessed. The effect of plastisphere on the biogeochemical cycles of blue carbon habitats has to be studied. The marine microbial communities in water columns are different from those on plastic debris. Though sea is a major contributor toward marine plastic debris, majority of the studies on marine plastisphere are conducted in temperate and subtropical marine atmospheres. Keeping this in mind, further studies should be done regarding finding solutions to questions such as: What is the gravity of difference of microbial communities in water columns and that of plastic debris in SEA water compared with temperate and subtropical waters (Abreo et al., 2018; Bednarz & Ludwig, 1997; Dobrucka, 2019).

## 30.9 Conclusions

This is noteworthy to observe different negative impacts and harms of BP on marine environment to our society, which includes the release of additives, microplastics, degradation products, and various other pollutants. These marine bioplastics can allow us to initiate a change in current material cycles of conventional plastics before their decomposition, change microbial communities. These bioplastic materials are also have great impact on various biogeochemical activities in the surrounding environment because of their biodegradability feature. However, the premise of these negative effects is the wide use and unscientific disposal of BP is not an established fact. Enhancing the rate of recycle of few plastic products and usage of long-lasting components such as metal or ceramics can decrease the proportion of disposable plastics, which in turn reduces the usage of BP. It is also an important fact that all the adverse effects are not yet confirmed. This is because of the limited studies about the ecological risks of BPs in the marine environment. It is worth noting that some of the imperfections of BPs can be conquered. If waste management is improved, it can largely have a positive effect on the negative impact of BPs on marine environment. It is also to be noted that BPs show a lot of advantages when it comes to medical applications. BP is also a good option for fishing gears, but most of the times it is lost to the marine environment. It is not an easy task for human beings to reduce the plastic pollutants getting into marine environment.

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